

1. EEG Processing

EEG Acquisition

OpenBCI (8-ch) | PhysioNet (64-ch)

Raw Signal

Preprocessing

Bandpass (8–30 Hz) → Fixed-Length Epochs

Comparison

Baseline Classifier

OVR-CSP + LDA

Filtered Epochs

2. EEGTransformer Model

Convolutional Front-End

Temporal → Spatial → Separable

Patch Embeddings

Spatiotemporal Tokenization

Transformer Encoder

Multi-Head Self-Attention

Residual Fusion

CNN Features $\oplus$ Global Context

Residual

Predicted Class
& Confidence

3. Closed-Loop RL Control

State Representation s_t

Kinematics $\oplus$ Classifier Output

DQN Agent

Transformer / CNN+LSTM Q-Net

Action a_t

Robotic Environment

SO-101 Arm Manipulation

Next State s_{t+1}
& Reward r_t

Hardware Integration

BrainFlow → OpenBCI
Serial Protocol → SO-101
(Feetech STS3215)

4-class MI → 4 commands: left, right, up, down